

EXCELLIA TRADE

Technical Documentation

Cloud-Native Fintech Solution

Property	Value
Document Title	Excellia Trade - Technical Architecture & Solution Overview
Version	1.0
Date	2026
Classification	Internal - Confidential
Department	Product Development & Engineering

Table of Contents

- 1. Executive Summary
- 2. Product Overview
- 3. Technical Architecture
- 4. Cloud Infrastructure
- 5. Core Modules & Functionality
- 6. Fintech Capabilities
- 7. Security & Compliance
- 8. Integration & Deployment
- 9. Performance & Scalability
- 10. Technical Specifications
- 11. API Documentation
- 12. Glossary

1. Executive Summary

1.1 Purpose

Excellia Trade is a comprehensive, cloud-native solution designed to manage and secure international banking operations. The platform provides end-to-end management of trade finance operations including documentary credits, documentary collections, international guarantees, financing, and SWIFT message processing.

1.2 Key Value Propositions

- Complete Functional Coverage: End-to-end trade finance operations management
- Optimized User Experience: Three-tier distributed architecture
- Flexible Integration: SaaS or on-premise deployment options
- Cloud-First Design: Modern, scalable cloud infrastructure
- Regulatory Compliance: Built-in compliance with international banking regulations

1.3 Target Audience

- International banks and financial institutions
- Trade finance departments
- Corporate treasury teams
- Financial operations managers
- Compliance officers

2. Product Overview

2.1 Core Capabilities

Module	Description	Key Features
Documentary Credits	Management of letters of credit	Issuance, amendments, advices, presentations, settlements
Documentary Collections	Collection processing	Outward/inward collections, clean collections, documentary collections
International Guarantees	Guarantee management	Issuance, amendments, claims handling, expiry management
Financing	Trade finance operations	Pre-shipment/post-shipment financing, discounting, factoring
SWIFT Processing	SWIFT message handling	MT/MX messages, ISO 20022 compliance, real-time tracking

3. Technical Architecture

3.1 Architectural Principles

- Microservices Architecture: Domain-driven design, containerized services, RESTful APIs, event-driven communication
- Cloud-Native Design: Kubernetes orchestration, auto-scaling capabilities, service mesh integration, multi-cloud compatibility
- Multi-Tier Structure: Client tier (UI/UX), Agency tier (branch operations), Head Office tier (central management)

3.2 Technology Stack

Layer	Technology	Purpose
Frontend	React.js / Angular	User interface
Backend	Java / Spring Boot / Python	Business logic
Database	PostgreSQL / MongoDB	Data storage
Messaging	Apache Kafka	Event streaming
Cache	Redis	Performance optimization
Search	Elasticsearch	Document retrieval
Orchestration	Kubernetes	Container management
API Gateway	Kong / NGINX	API management
Monitoring	Prometheus / Grafana	Observability

4. Cloud Infrastructure

4.1 Cloud Deployment Models

Model	Description	Best For
SaaS	Fully managed cloud service	Rapid deployment, reduced operational overhead
On-Premise	Self-hosted within organization	Security requirements, data sovereignty
Hybrid	Combination of both	Phased migration, specific compliance needs

4.2 Cloud Providers Supported

Provider	Services Used	Region Support
AWS	EKS, RDS, S3, ElastiCache	Global
Azure	AKS, Azure SQL, Blob Storage	Global
GCP	GKE, Cloud SQL, Cloud Storage	Global

5. Core Modules & Functionality

5.1 Documentary Credit Management

Features:

- Credit issuance and amendments
- LC types: revocable, irrevocable, confirmed
- Presentation handling
- Discrepancy management
- Payment settlement

Workflow: Application → Issuance → Advice → Presentation → Examination → Settlement

5.2 Documentary Collections

- Clean collections
- Documentary collections
- Outward collections
- Inward collections
- Protest handling

5.3 International Guarantees

- Guarantee issuance
- Bid bonds
- Performance guarantees
- Advance payment guarantees
- Claims management
- Expiry monitoring

6. Fintech Capabilities

6.1 Digital Transformation Features

Capability	Description	Benefit
Digital Onboarding	Electronic KYC/AML verification	Faster client onboarding
Smart Document Processing	AI-powered document extraction	Reduced manual errors
Real-Time Analytics	Live dashboards and KPIs	Better decision making
Mobile Accessibility	Mobile-responsive interface	Anywhere, anytime access
API-First Design	Full API coverage	Easy integration

6.2 AI & Machine Learning Integration

- Document Processing: OCR, Intelligent Document Recognition, Automated Data Extraction, Classification & Categorization
- Risk Management: Fraud Detection, Anomaly Detection, Predictive Analytics
- Operational Efficiency: Intelligent Automation, Smart Workflows, Pattern Recognition

7. Security & Compliance

7.1 Security Architecture

Layer	Security Controls
Network	VPC, firewalls, WAF, DDoS protection
Application	JWT/OAuth2, RBAC, MFA, encryption
Data	AES-256 encryption, TLS 1.3, data masking
Infrastructure	Security patches, vulnerability scanning, SIEM

7.2 Compliance Standards

- SOC 2: Security, availability, processing integrity, confidentiality, privacy
- ISO 27001: Information security management
- PCI DSS: Payment card industry data security (if applicable)
- GDPR: Data protection and privacy
- SWIFT CSP: Customer Security Programme compliance
- KYC/AML: Know Your Customer / Anti-Money Laundering

8. Integration & Deployment

8.1 Integration Capabilities

- Banking Systems: Core banking integration
- Legacy Systems: Mainframe and legacy system connectivity
- Third-Party Services: External service integration
- API Gateway: REST APIs, Webhooks, Event Streams

8.2 Deployment Options

Option	Description	Time to Deploy
SaaS	Fully managed cloud	1-2 weeks
On-Premise	Self-hosted	4-8 weeks
Hybrid	Mixed deployment	3-6 weeks

9. Performance & Scalability

9.1 Performance Metrics

Metric	Target	Description
Response Time	< 200ms	Average API response time
Throughput	10,000+ TPS	Transactions per second
Availability	99.99%	System uptime
Recovery Time	< 15 min	RTO (Recovery Time Objective)
Data Consistency	< 1 sec	Replication lag

9.2 Scalability Features

- Horizontal Scaling: Add more instances as needed
- Vertical Scaling: Upgrade resources as needed
- Auto-Scaling: Dynamic resource allocation
- Sharding: Data distribution across nodes
- Caching: In-memory data acceleration
- CDN: Content delivery network

10. Technical Specifications

10.1 Software Requirements

Component	Version	Description
Database	PostgreSQL 14+	Primary database
Cache	Redis 6+	In-memory caching
Message Queue	Apache Kafka 2.8+	Event streaming
Search Engine	Elasticsearch 7.x	Document search
Container Runtime	Docker 20.10+	Containerization
Orchestration	Kubernetes 1.24+	Container orchestration
API Gateway	Kong 2.8+	API management

10.2 Hardware Requirements

Environment	CPU	Memory	Storage	Network
Development	4 cores	8 GB	100 GB	1 Gbps
Staging	8 cores	16 GB	500 GB	1 Gbps
Production	16+ cores	32+ GB	1+ TB	10 Gbps

11. API Documentation

11.1 Base URL

https://api.excellia.com/v1/

11.2 Authentication

Authorization: Bearer

11.3 Core APIs

API	Method	Endpoint	Description
Create LC	POST	/credits	Create new letter of credit
View LC	GET	/credits/{id}	View LC details
Amend LC	PUT	/credits/{id}	Amend existing LC
Create Collection	POST	/collections	Create new collection
Create Guarantee	POST	/guarantees	Create new guarantee
Process SWIFT	POST	/swift/process	Process SWIFT message
Search	GET	/search	Search operations
Reports	GET	/reports/{type}	Generate reports

12. Glossary

Term	Definition
Documentary Credit (LC)	A commitment by a bank to pay a seller on behalf of a buyer
SWIFT	Society for Worldwide Interbank Financial Telecommunication
ISO 20022	International standard for financial messaging
Fintech	Financial technology
SaaS	Software as a Service
RAG	Retrieval-Augmented Generation
KYC	Know Your Customer
AML	Anti-Money Laundering
MT Message	SWIFT message type (legacy)
MX Message	ISO 20022 message format
LC	Letter of Credit
RTO	Recovery Time Objective
RPO	Recovery Point Objective
RBAC	Role-Based Access Control
MFA	Multi-Factor Authentication
TPS	Transactions Per Second
CDN	Content Delivery Network

© 2026 Excellia RAG. All rights reserved.

This document contains proprietary information and is confidential.

Unauthorized reproduction or distribution is prohibited.